

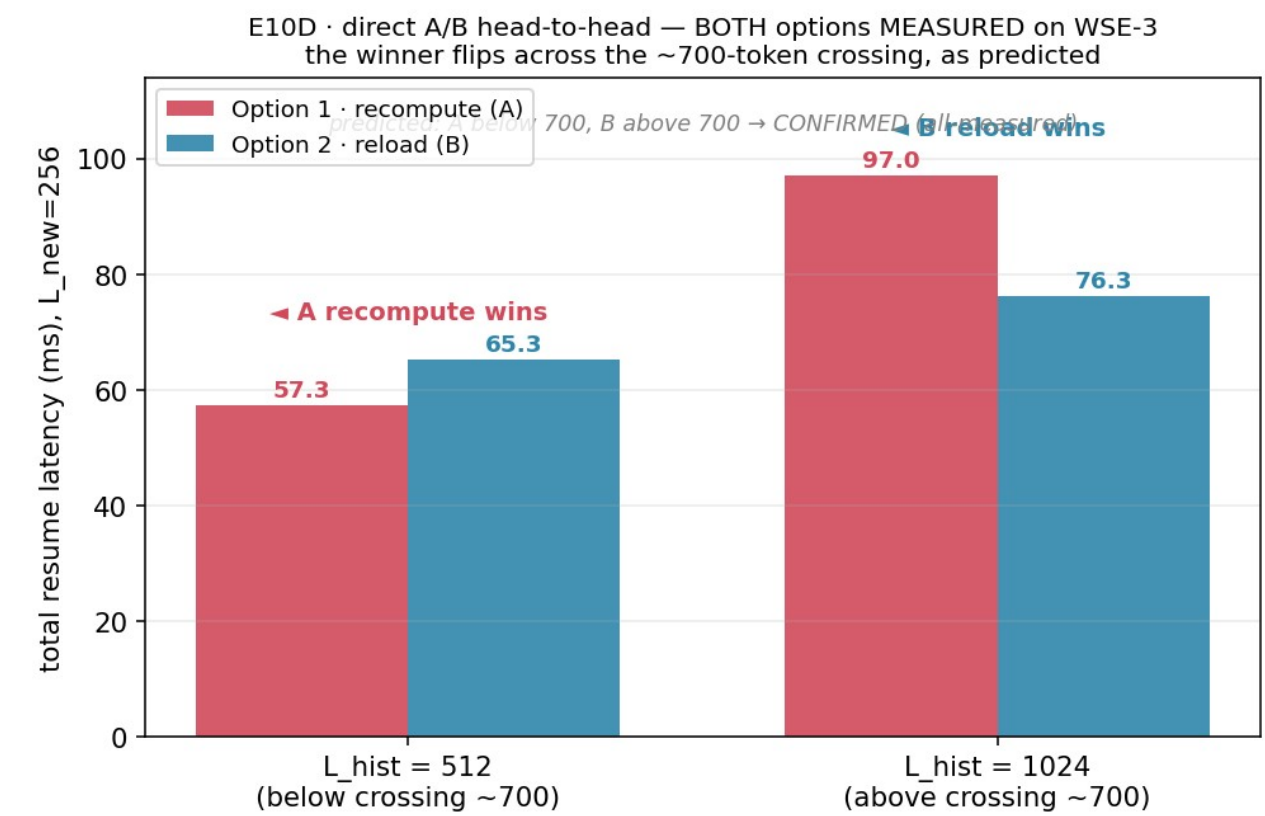
WEEKLY COLLABORATOR DISCUSSION

On-Chip KV Offload as a Storage Tier

Two multi-row storage designs measured on real WSE-3; the on-chip tier now sits under both prior resume options.

MODEL	EVIDENCE	SCOPE	DATE
Qwen3-1.7B geometry	81 CS-3 runs, 0-cycle spread	M3 idle-PE tier	2026-08-24

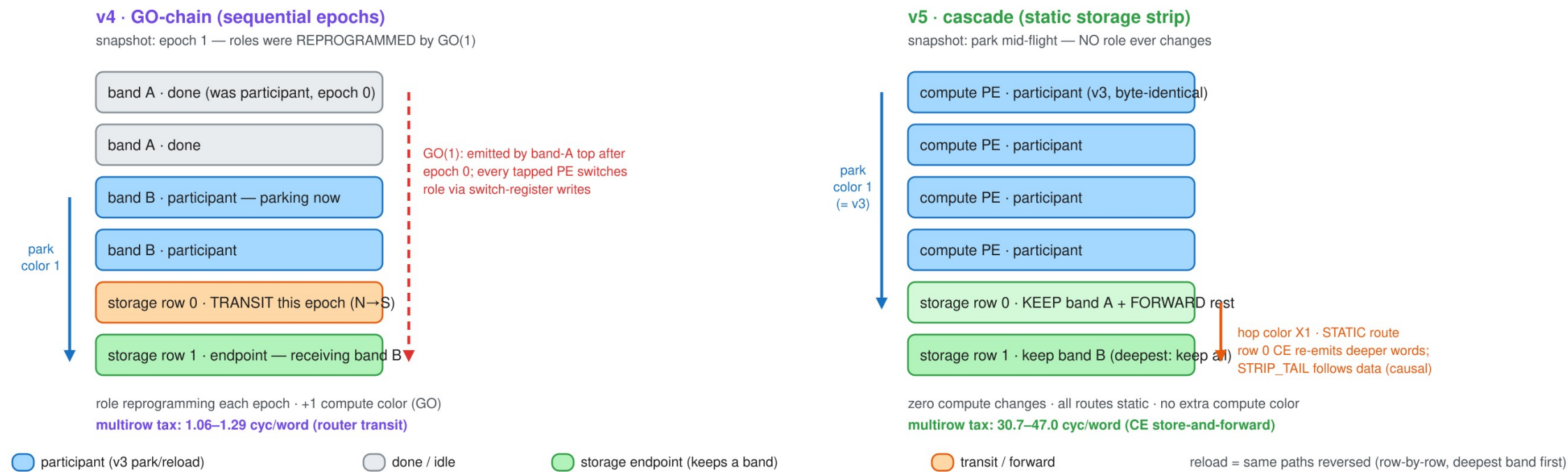
M2 ended at recompute-vs-host with a 744-token crossing; M3 asks if idle PEs beat both



Last time (M2 E10/E10D, all device-measured): below H*=744 recompute wins, above it host reload wins — but reload rides shared edge IO and the host. M3's question: can idle on-chip PEs hold parked KV as a middle tier? This week: build the multi-row storage strip both candidate ways, measure both end-to-end, and price the tier.

v4 reprograms compute roles per epoch; v5 keeps compute byte-identical and cascades inside the strip

Two multi-row designs (R=2 shown) — snapshot: parking the second half of the column



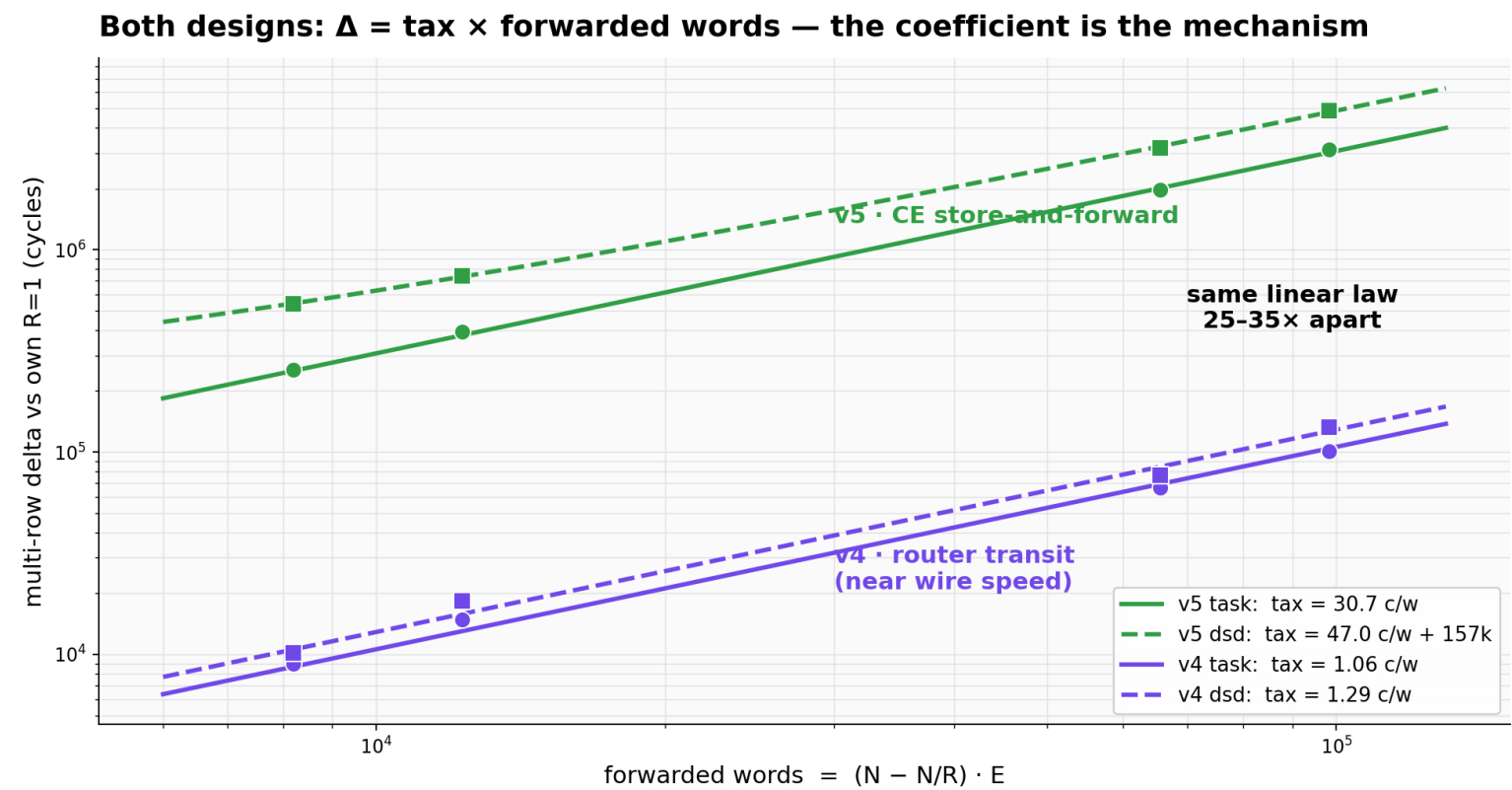
Both share the proven single-row protocol on one borrowed color. v4: bands park in sequential epochs, chained fully on-chip by a GO color; every tapped PE rewrites its switch positions per epoch. v5: row 0 keeps the nearest band and store-and-forwards deeper words on static per-hop colors; a STRIP_TAIL word rides behind the data as the causal end-of-stream. Reload returns over the same paths reversed.

RESULTS · 81 RUNS, EVERY CELL N=3 WITH 0-CYCLE SPREAD

Full park+reload cycle (cycles): v5 wins single-row, v4 wins every multi-row cell

CONFIG (N=256)	V3 SINGLE-ROW	V4 GO-CHAIN	V5 CASCADE
task E=64 · R=1	1,388,470	1,535,920	1,420,244
task E=64 · R=2	—	1,544,946	1,675,327
task E=64 · R=4	—	1,550,855	1,812,526
dsd E=512 · R=1	7,350,234	8,542,434	7,482,812
dsd E=512 · R=2	—	8,619,349	10,669,498
dsd E=512 · R=4	—	8,675,026	12,343,610

Both designs obey $\Delta = \text{tax} \times \text{forwarded words}$ — the tax is the mechanism, 25-35x apart



Delta vs each design's own $R=1$ baseline collapses on the same axis, $\text{fwd_words} = (N - N/R) \cdot E$. v4's tax is router-level transit (1.06/1.29 cyc/word, near wire speed); v5's is a CE data-task per forwarded word (30.7/47.0). The v5 dsd fit needs one extra constant ($\sim 157\text{k}$ cycles): strip rows emit their own block before relaying, which fast owners expose.

The equations separate endpoint work from movement

39 CS-3 anchors; max error 0.89% / 0.83% / 0.23% (v3/v5/v4). Pairs are task | owner-side bulk-DSD reload.

MEASURED CYCLE MODELS

One baseline plus one movement term

$$\text{fwd} = (N - N/R) \times E$$

$$\begin{aligned} v3 = & (245.4N + 326) \\ & + (E-4)N(86.3 \mid 56.0) + 4(D-1) \end{aligned}$$

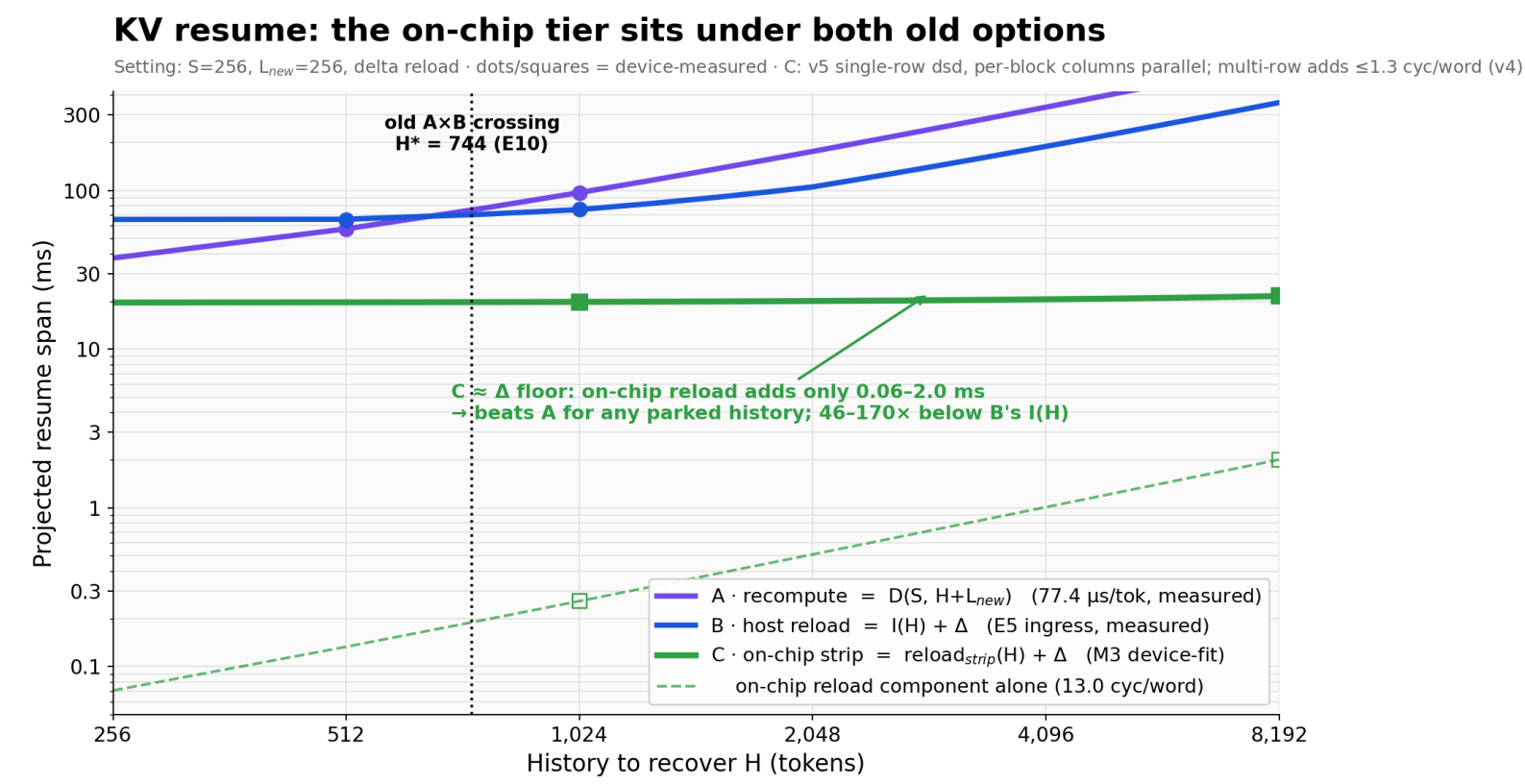
$$\begin{aligned} v4 = & (61,058 \mid 78,676) \\ & + (E-4)N(96.00 \mid 65.08) \\ & + (1.06 \mid 1.29)\text{fwd} \end{aligned}$$

$$\begin{aligned} v5 = & (70,561 \mid 70,076) \\ & + (E-4)N(87.87 \mid 57.00) \\ & + (30.7 \mid 47.0)\text{fwd} + 157k^* \end{aligned}$$

HOW TO READ THE COEFFICIENTS

- 01** N/E/R/D = compute PEs, words/PE, storage rows, distance; each design owns R=1.
- 02** Floor + marginal = launch/control + endpoint work; task | DSD isolates owner tasks.
- 03** fwd separates router (v4: 1.06 | 1.29) from CE store-and-forward (v5: 30.7 | 47.0).

The on-chip tier sits under both old options: resume is now delta-dominated



Same axes and setting as the M2 boundary figure, plus measured tier C: on-chip strip reload adds only 0.06-2.0 ms (13.0 cyc/word), 46-170x below host ingress $I(H)$, flattening resume onto the ~ 20 ms L_{new} delta floor. The old 744-token crossing becomes a fallback choice for un-parked history. Next lever for resume latency is the delta/restart path, not the transfer.